Bioanalytical Methods for the Food Industry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Bioanalytical Methods for the Food Industry (A06317)

Short Title:	Bioanalytical Methods Food Ind		
Faculty	Science and Computing		
Department	Science		
Cluster	Food Science		
Author	LCOFFEY		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module aims to provide an understanding of techniques required in undertaking detailed analysis of food ingredients, processes and products, with a focus on food safety.

Programmes

BIOL-0025	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	2 / 3 / M
BIOL-0025	BSc in Food Science with Business (WD_SFDSC_D)	2 / 4 / M

Indicative Content

- Traditional microbiological methods for the analysis of food ingredients, processes and products
- Rapid microbiological and immunological methods for the identification of food borne pathogens/ indicator organisms; this will include giving examples and explaining the principles behind chromogenic media
- The principles and applications of ELISA-based assays for relevant testing
- The use of enzymes in food production/analysis
- Rapid hygiene testing methods relevant to the food industry
- Methods for food allergen analysis in food
- The concept of genetically modified foods; their applications, potential and their detection
- PCR as a DNA-based testing method relevant to the food industry

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare and contrast traditional microbiological methods of food analysis with modern rapid kits, techniques and media.
2. Identify and apply the appropriate modern bioanalytical technique/kit for the chosen substance/ parameter.
3. Describe the various bioanalytical methods available for relevant testing such as hygiene, allergen and pathogen detection from foods/food associated surfaces.
4. Describe the principles of ELISA and select the appropriate format for the relevant sample.
5. Describe the concept of GM foods and the associated issues.
6. Identify the basis of a DNA-based testing method relevant to the food industry.
7. Demonstrate proficiency in practical work focused on indicative content.

Learning and Teaching Methods

- Lectures.
- Self-directed study.
- Laboratory-based practicals.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6
Continuous Assessment	50%	
Practical	50%	7

Assessment Criteria

<40%: Very little understanding of techniques covered.
40%-49%: Has a basic understanding of techniques covered.
50%-59%: Good knowledge of methods and has the ability to perform such methods in a laboratory setting.
60%-69%: Good knowledge of methods and has the ability to perform such methods in a laboratory setting, also understands the principles behind such bioanalytical methods with the potential to troubleshoot.
70%-100%: Displays an in-depth knowledge of all relevant current bioanalytical techniques and displays excellent ability in identifying the most appropriate technique, also performing lab experiments designed with such bioanalytical tools.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Lab	24	
Independent Learning	99	

Essential Material

- Shaw, I. _Food Safety: The science of keeping food safe_. UK: Wiley Blackwell, 2013.

Supplementary Material

- D'Mello, J.P. _Food Safety: contaminants and toxins_. Cambridge, MA: Oxford, 2003.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab